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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
19/281,727	07/27/2025	Shuai WANG	00144.1675.00US	4218

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EXAMINER
RINK, RYAN J

ART UNIT	PAPER NUMBER
3619	

NOTIFICATION DATE	DELIVERY MODE
09/04/2026	ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

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Office Action Summary	Application No. 19/281,727	Applicant(s) WANG et al.	
	Examiner RYAN J RINK	Art Unit 3619	AIA (First Inventor to File) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1)☑ Responsive to communication(s) filed on 07/27/2025.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2a)☐ This action is **FINAL.** 2b) ☑ This action is non-final.

3)☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

4)☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

5) ☑ Claim(s) 1-20 is/are pending in the application.

5a) Of the above claim(s) _____ is/are withdrawn from consideration.

6) ☐ Claim(s) _____ is/are allowed.

7) ☑ Claim(s) 1,4-5,8,11,13,16-17 and 20 is/are rejected.

8) ☑ Claim(s) 2-3,6-7,9-10,14-15 and 17-19 is/are objected to.

9) ☐ Claim(s) _____ are subject to restriction and/or election requirement

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to **PPHfeedback@uspto.gov.**

Application Papers

10)☐ The specification is objected to by the Examiner.

11)☑ The drawing(s) filed on 07/27/2025 is/are: a)☑ accepted or b)☐ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

12)☑ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a)☑ All b)☐ Some** c)☐ None of the:

1.☑ Certified copies of the priority documents have been received.

2.☐ Certified copies of the priority documents have been received in Application No. _____.

3.☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☑ Notice of References Cited (PTO-892)

2) ☑ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b) Paper No(s)/Mail Date _____.

3) ☐ Interview Summary (PTO-413) Paper No(s)/Mail Date _____.

4) ☐ Other: _____.

Notice of Pre-AIA or AIA Status

The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

DETAILED ACTION

This is a non-final Office Action on the merits. Claims *** are currently pending and are addressed below.

Priority

Acknowledgment is made of applicant's claim for priority application No. *** filed on ***.

Information Disclosure Statement

The information disclosure statement (IDS) submitted on *** is being considered by the examiner.

Claim Rejections - 35 USC § 102

In the event the determination of the status of the application as subject to AIA 35 U.S.C. 102 and 103 (or as subject to pre-AIA 35 U.S.C. 102 and 103) is incorrect, any correction of the statutory basis for the rejection will not be considered a new ground of rejection if the prior art relied upon, and the rationale supporting the rejection, would be the same under either status.

The following is a quotation of the appropriate paragraphs of 35 U.S.C. 102 that form the basis for the rejections under this section made in this Office action:

A person shall be entitled to a patent unless –

(a)(1) the claimed invention was patented, described in a printed publication, or in public use, on sale or otherwise available to the public before the effective filing date of the claimed invention.
(a)(2) the claimed invention was described in a patent issued under section 151, or in an application for patent published or deemed published under section 122(b), in which the patent or application, as the case may be, names another inventor and was effectively filed before the effective filing date of the claimed invention.

Claim(s) 1, 5, 8, 12, 13, 17, and 20 is/are rejected under 35 U.S.C. 102(a)(1) as being anticipated by Tsai et al. (Adaptive Motion Control of a Terrain-Adaptive Self-Balancing Leg-Wheeled Mobile Robot over Rough Terrain).

Regarding claim 1:

Tsai teaches a balance control method for a wheel-legged robot **(see at least abstract, Fig. 1)**, the method being executed by a computer device **(Fig. 2)**, the wheel-legged robot comprising: a moving wheel, n links, and n rotating joints, the moving wheel being connected to a first link of the n links through a first rotating joint of the n rotating joints, and the n links being connected in series through n-1 rotating joints other than the first rotating joint, where n being a positive integer greater than 1 **(see at least Fig. 1)**; the method comprising:

acquiring a state quantity of the wheel-legged robot at a first moment, the state quantity at the first moment indicating a motion state of the wheel-legged robot at the first moment **(see at least IIIA, Dynamic modeling, IIIB. Stable trajectory tracking control)**;

determining dynamics model parameters according to a dynamics equation of the wheel-legged robot and the state quantity at the first moment, the dynamics model parameters being configured to define a mapping relationship between an angular

acceleration at the first moment and a rotation torque at a second moment, the angular acceleration at the first moment comprising angular accelerations of the n links and an angular acceleration of the moving wheel, and the rotation torque at the second moment comprising rotation torques of the n rotating joints at the second moment **(see at least IIIA, Dynamic modeling)**;

establishing a sliding surface according to the state quantity at the first moment, the state quantity of the wheel-legged robot approaching a stable value on the sliding surface **(see at least IIIB. Stable trajectory tracking control)**;

calculating the rotation torques of the n rotating joints according to the sliding surface and the dynamics model parameters **("Cθ and Cy stand for torque control signals", eq. 4 and 5)**; and

controlling the n rotating joints according to the rotation torques of the n rotating joints at the second moment **(see at least B. Stable Trajectory Tracking Control)**.

Regarding claim 5 and 17:

Tsai further teaches wherein the sliding surface comprises n sliding surfaces, and the n sliding surfaces are configured to constrain the rotation torques of the n rotating joints;

the establishing a sliding surface according to the state quantity at the first moment comprises:

determining at least two sliding mode parameters for an ith sliding surface of the n sliding surfaces, i being a positive integer less than or equal to n; and

establishing the ith sliding surface according to the at least two sliding mode parameters and the state quantity at the first moment **(see at least III B. Stable Trajectory Tracking Control).**

Regarding claim 8 and 20:

Tsai further teaches wherein the controlling the n rotating joints according to the rotation torques of the n rotating joints comprises:

for a rotating joint of the n rotating joints, controlling a rotating motor corresponding to the rotating joint to rotate according to the rotation torque of the rotating joint **(see at least III A, torque control signals, 2A, drive motors)**.

Regarding claims 12 and 13, Tsai teaches a balance control apparatus and computer readable medium for performing the method as in claim 1 above.

Claim Rejections - 35 USC § 103

In the event the determination of the status of the application as subject to AIA 35 U.S.C. 102 and 103 (or as subject to pre-AIA 35 U.S.C. 102 and 103) is incorrect, any correction of the statutory basis for the rejection will not be considered a new ground of rejection if the prior art relied upon, and the rationale supporting the rejection, would be the same under either status.

The following is a quotation of 35 U.S.C. 103 which forms the basis for all obviousness rejections set forth in this Office action:

> A patent for a claimed invention may not be obtained, notwithstanding that the claimed invention is not identically disclosed as set forth in section 102 of this title, if the differences between the claimed invention and the prior art are such that the claimed invention as a whole

would have been obvious before the effective filing date of the claimed invention to a person having ordinary skill in the art to which the claimed invention pertains. Patentability shall not be negated by the manner in which the invention was made.

Claims 4 and 16 are rejected under 35 U.S.C. 103 as being unpatentable over Tsai as in claim 1 above, in view of Cui et al. (Modeling and Control of a Wheeled Biped Robot).

Regarding claim 4 and 16:

Tsai further teaches wherein the dynamics equation of the robot is obtained by abstracting the wheel-legged robot into an n-level inverted pendulum model **(see at least IV A, Impedance control)**

Tsai is silent as to performing derivation based on an Euler-Lagrange equation.

Cui teaches a system and method of controlling a wheel legged mobile robot, including modelling the robotic system utilizing an inverted pendulum model and performing derivation based on an Euler-Lagrange equation **(see at least abstract, section 3.2).**

It would have been obvious to one of ordinary skill in the art at the time of filing of the invention to modify the inverted pendulum model for controlling the wheel-legged mobile robot as taught by Tsai with the well-known Euler-Lagrange equation for modeling an inverted pendulum as taught by Cui in order to provide the expected result of providing a systematic method to handle complex, multi-variable, and nonlinear systems without needing to track internal constraint forces.

Claim Rejections - 35 USC § 103

Claim 11 and are rejected under 35 U.S.C. 103 as being unpatentable over Tsai.

Regarding claim 11:

Tsai teaches the limitations as in claim 1 above.

Tsai further teaches wherein the state quantity comprises: the deflection angles of the n links, the angular velocities of the n links, and the angular velocity of the moving wheel **(see at least Section III. BACKSTEPPING SLIDING-MODE MOTION CONTROL).**

Tsai further is silent as to the particular sensors used for determining the state quantities.

Tsai does further teach at least the wheels including encoders.

It would have been obvious to one of ordinary skill in the art at the time the invention was filed to modify the robot control system and method as taught by Tsai with any known sensor types for determining states of the robot, including conventional sensor types as IMUs and encoders as a matter of design choice.

Allowable Subject Matter

Claims objected to as being dependent upon a rejected base claim, but would be allowable if rewritten in independent form including all of the limitations of the base claim and any intervening claims.

Conclusion

The prior art made of record and not relied upon is considered pertinent to applicant's disclosure.

Any inquiry concerning this communication or earlier communications from the examiner should be directed to RYAN J RINK whose telephone number is (571)272-4863. The examiner can normally be reached M-F 8-5.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at http://www.uspto.gov/interviewpractice.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Anna Momper can be reached on (571) 270-5788. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

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Representative, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

/Ryan Rink/
Primary Examiner, Art Unit 3619